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Inventor(s)	Kim; Jiwon et al.

Semiconductor device having peripheral circuit areas at both sides of substrate and data storage system including the same

Abstract

A semiconductor device including a cell area including a first substrate, gate electrodes on the first substrate, a channel structure extending through the gate electrodes, cell contact plugs, a through contact plug, and first bonding pads, the first peripheral circuit area including second bonding pads on the first bonding pads; a second peripheral circuit area connected to the first peripheral circuit area; and a second substrate between the first peripheral circuit area and the second peripheral circuit area, the second substrate including a first surface in the first peripheral circuit area and a second surface in the second peripheral circuit area, wherein the second peripheral circuit area includes a device on the second surface, and a through electrode extending vertically through the second substrate and connected to the first peripheral circuit area.

Inventors: Kim; Jiwon (Seoul, KR), Ahn; Jaeho (Seoul, KR), Hwang; Sungmin (Hwaseong-si, KR), Lim; Joonsung (Seongnam-si, KR), Sung; Sukkang (Seongnam-si, KR)

Applicant: SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)

Family ID: 1000008782596

Assignee: Samsung Electronics Co., Ltd. (Suwon-si, KR)

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Primary Examiner: Ojha; Ajay

Assistant Examiner: Chiu; Tsz K

Attorney, Agent or Firm: Fish & Richardson P.C.

Background/Summary

CROSS-REFERENCE TO THE RELATED APPLICATION

(1) Korean Patent Application No. 10-2020-0141233, filed on Oct. 28, 2020, in the Korean Intellectual Property Office, and entitled: “Semiconductor Device Having Peripheral Circuit Areas at Both Sides of Substrate and Data Storage System Including the Same,” is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

(2) Embodiments relate to a semiconductor device having peripheral circuit areas at both sides of a substrate and a data storage system including the same.

2. Description of the Related Art

(3) A 3-dimensional nonvolatile memory device having a multi-stack structure has been considered for lightness, thinness, simplification, miniaturization, and high integration of electronic products. Such a nonvolatile memory device may include a cell area, and a peripheral circuit area connected to the cell area.

SUMMARY

(4) The embodiments may be realized by providing a semiconductor device including a cell area including a first substrate, gate electrodes on the first substrate and spaced apart from one another in a vertical direction, a channel structure extending vertically through the gate electrodes, cell contact plugs connected to the gate electrodes and extending in the vertical direction, a through contact plug connected to the first substrate and extending in the vertical direction, and first bonding pads electrically connected to the channel structure, the cell contact plugs and the through contact plug; a first peripheral circuit area connected to the cell area on the cell area, the first peripheral circuit area including second bonding pads on the first bonding pads; a second peripheral circuit area connected to the first peripheral circuit area; and a second substrate between the first peripheral circuit area and the second peripheral circuit area, the second substrate including a first surface in the first peripheral circuit area and a second surface in the second peripheral circuit area, wherein the second peripheral circuit area includes a device on the second surface, and a through electrode extending vertically through the second substrate and connected to the first peripheral circuit area.

(5) The embodiments may be realized by providing a semiconductor device including a cell area including a first substrate, gate electrodes on a first surface of the first substrate and spaced apart from one another in a vertical direction, a channel structure extending vertically through the gate electrodes, cell contact plugs connected to the gate electrodes and extending in the vertical direction, a through contact plug connected to the first substrate and extending in the vertical direction, and first bonding pads electrically connected to the channel structure, the cell contact plugs, and the through contact plug; a first peripheral circuit area under the cell area and connected to the cell area, the first peripheral circuit area including second bonding pads on the first bonding pads; and a second peripheral circuit area connected to the cell area on the cell area, wherein the first substrate is between the cell area and the second peripheral circuit area, the first substrate including the first surface in the cell area and a second surface in the second peripheral circuit area, and wherein the second peripheral circuit area includes a device on the second surface, and a through electrode extending vertically through the first substrate such that the through electrode is electrically connected to the cell area.

(6) The embodiments may be realized by providing a semiconductor device including a cell area including a first substrate, gate electrodes on the first substrate and spaced apart from one another in a vertical direction, a channel structure extending vertically through the gate electrodes, cell contact plugs connected to the gate electrodes and extending in the vertical direction, a through contact plug connected to the first substrate and extending in the vertical direction, and first bonding pads electrically connected to the channel structure, the cell contact plugs, and the through contact plug; a first peripheral circuit area on the cell area and connected to the cell area, the first peripheral circuit area including second bonding pads on the first bonding pads, respectively; a second peripheral circuit area on the first peripheral circuit area and connected to the first peripheral circuit area; a semiconductor storage device including a second substrate between the first peripheral circuit area and the second peripheral circuit area and including a first surface in the first peripheral circuit area and a second surface in the second peripheral circuit area, and an input/output pad electrically connected to at least one of the first peripheral circuit area or the second peripheral circuit area; and a controller electrically connected to the semiconductor storage device through the input/output pad such that the controller controls the semiconductor storage device, wherein the second peripheral circuit area includes a device on the second surface, and a

through electrode extending vertically through the second substrate such that the through electrode is electrically connected to the first peripheral circuit area.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Features will be apparent to those of skill in the art by describing in detail exemplary embodiments with reference to the attached drawings in which:
- (2) FIG. 1 is a diagram of a memory system according to an exemplary embodiment.
- (3) FIG. 2 is a memory card according to an exemplary embodiment.
- (4) FIG. 3 is a cross-sectional view of the semiconductor package shown in FIG. 2 taken along line I-I'.
- (5) FIG. 4 is a cross-sectional view of a semiconductor device according to an exemplary embodiment.
- (6) FIG. 5 is an enlarged view of the semiconductor device shown in FIG. 4.
- (7) FIGS. 6 to 16 are cross-sectional views of stages in a method of manufacturing a semiconductor device shown in FIG. 4 according to an exemplary embodiment.
- (8) FIG. 17 is a cross-sectional view of a semiconductor device according to an exemplary embodiment.
- (9) FIGS. 18 to 21 are cross-sectional views of stages in a method of manufacturing a semiconductor device shown in FIG. 17 according to an exemplary example embodiment.
- (10) FIGS. 22 to 29 are cross-sectional views of semiconductor devices according to exemplary embodiments.

DETAILED DESCRIPTION

- (11) FIG. 1 is a diagram of a memory system according to an exemplary embodiment.
- (12) Referring to FIG. 1, a memory system **1000** may include a memory device **1100** and a controller **1200** connected to a pad **1101** of the memory device **1100**. The memory device **1100** may include a cell area **1100S** and a peripheral circuit area **1100F**.
- (13) The cell area **1100S** may include a plurality of cell strings CSTR each including memory cell transistors MCT connected to one another in series, and a first upper transistor UT1, a second upper transistor UT2, a first lower transistor LT1, and a second lower transistor LT2, which are connected to opposite ends of the memory cell transistors MCT. The plurality of cell strings CSTR may be connected to corresponding ones of bit lines BL in parallel, respectively. The plurality of cell strings CSTR may be connected to a common source line CSL in common. In an implementation, a plurality of cell strings CSTR may be between a plurality of bit lines BL and a single common source line CSL.
- (14) The memory cell transistors MCT, which are connected to one another in series, may be controlled by word lines WL for selecting cell strings CSTR. Each of the memory cell transistors MCT may include a data storage element. Gate electrodes of the memory cell transistors MCT spaced apart from the common source line CSL by the same distance may be connected to one of the word lines WL in common and, as such, may be in an equipotential state. In an implementation, even when the gate electrodes of the memory cell transistors MCT are spaced apart from the common source line CSL by the same distance, the gate electrodes, which are in different rows or columns, may be independently controlled.
- (15) The first lower transistor LT1 and the second lower transistor LT2 may be ground selection transistors, respectively. The first lower transistor LT1 and the second lower transistor LT2 may be controlled by a first lower line LL1 and a second lower line LL2, respectively, and may be connected to the common source line CSL. The first upper transistor UT1 and the second upper transistor UT2 may be string selection transistors, respectively. The first upper transistor UT1 and

the second upper transistor UT2 may be controlled by a first upper line UL1 and a second upper line UL2, respectively, and may be connected to corresponding ones of the bit lines BL, respectively. In an embodiment, at least one dummy line or buffer line may be further disposed between an uppermost one of word lines WL and the first upper transistor UT1. At least one dummy line may also be disposed between a lowermost one of the word lines WL and the second lower transistor LT2. In the specification, the term “dummy” is used to represent a configuration which has a structure and a shape identical or similar to those of another constituent element, but is simply present as a pattern without performing a substantial function in a device.

(16) When a signal is applied to the first upper transistor UT1 and the second upper transistor UT2, which are string selection transistors, via the first upper line UL1 and the second upper line UL2, a signal applied to the corresponding bit line BL is transferred to the memory cell transistors MCT connected to one another in series and, as such, a data read or data write operation may be executed. In addition, when a predetermined erase voltage is applied through a substrate, a data erase operation for erasing data written in the memory cell transistors MCT may be executed. In an implementation, the cell area 1100S may include at least one dummy cell string CSTR electrically isolated from the bit lines BL.

(17) The peripheral circuit area 1100F may include a row decoder 1110, a page buffer 1120, and a logic circuit 1130. The row decoder 1110 may be connected to the word lines WL, the first upper transistor UT1, the second upper transistor UT2, the first lower transistor LT1, the second lower transistor LT2, and the common source line CSL. The page buffer 1120 may be connected to the bit lines BL via connection lines 1125. The logic circuit 1130 may be connected to the row decoder 1110 and the page buffer 1120, and may be connected to a controller 1200 via the pad 1101.

(18) The row decoder 1110 decodes an input address, thereby generating and transferring drive signals for the word lines WL. The row decoder 1110 may provide a word line voltage generated from a voltage generation circuit in the logic circuit 1130 under control of the logic circuit 1130 to a selected one of the word lines WL and an unselected one of the word lines WL.

(19) The page buffer 1120 may be connected to the cell area 1100S via a bit line BL and, as such, may read out information stored in a memory cell. The page buffer 1120 may temporarily store data to be stored in the memory cell or may sense data stored in the memory cell in accordance with an operation mode. The page buffer 1120 may include a column decoder and a sense amplifier. The column decoder may selectively activate the bit lines BL of the cell area 1100S. The sense amplifier may sense a voltage of the bit line BL selected by the column decoder in a read operation and, as such, may read out data stored in a selected memory cell.

(20) The logic circuit 1130 may control operation of the row decoder 1110 and operation of the page buffer 1120. The logic circuit 1130 may include a voltage generation circuit configured to generate voltages required for internal operations, for example, a program voltage, a read voltage, an erase voltage, etc., using an external voltage. The logic circuit 1130 may control a read operation, a write operation and/or an erase operation in response to control signals. In addition, the logic circuit 1130 may include an input/output circuit. In a program operation, the input/output circuit may receive data DATA input thereto, and may transfer the received data DATA to the page buffer 1120. In a read operation, the input/output circuit may receive data DATA from the page buffer 1120, and may output the received data DATA to the outside thereof. The logic circuit 1130 may be connected to the controller 1200 via a connection line 1135 and the pad 1101.

(21) The controller 1200 may include a processor 1210, a NAND controller 1220, and a host interface 1230. The processor 1210 performs control operations for data exchange of the NAND controller 1220. The NAND controller 1220 controls data exchange with the NAND controller 1220. The NAND controller 1220 may include a NAND interface 1221. The NAND interface 1221 interfaces with the memory device 1100 according to the exemplary embodiment of the disclosure. The host interface 1230 includes a data exchange protocol of a host connected to the memory system 1000.

(22) FIG. 2 is a memory card according to an exemplary embodiment. FIG. 3 is a cross-sectional view of the semiconductor package shown in FIG. 2 taken along line I-I'.

(23) Referring to FIG. 2, the memory system **2000** may be a solid state drive (SSD), a memory card, or a universal serial bus (USB) device. The memory system **2000** may include semiconductor packages **2003** (**2003a** and **2003b**), devices **2002** and **2004**, and a connector **2006** which are on a main substrate. In an implementation, each of the semiconductor packages **2003** (**2003a** and **2003b**) may include the memory device **1100** shown in FIG. 1. The devices **2002** and **2004** may correspond to the controller **1200**. In an implementation, each semiconductor package **2003** may include a package substrate **2100**, a memory stack **2200** on the package substrate **2100**, an adhesive layer **2300** between the package substrate **2100** and the memory stack **2200**, a wire **2400** electrically connecting the package substrate **2100** and the memory stack **2200**, and an encapsulator **2500** covering the package substrate **2100** and the memory stack **2200**.

(24) Further referring to FIG. 3, the package substrate **2100** may include a substrate pad **2130** and a wiring layer **2135**. The substrate pad **2130** may be at an upper surface of the package substrate **2100**, and may be connected to a chip pad **2210** via the wire **2400**. The wiring layer **2135** may be disposed in the package substrate **2100**, and may be electrically connected to the substrate pad **2130**.

(25) A plurality of memory devices **2200a** may be stacked on the package substrate **2100**. Each memory device **2200a** may correspond to a semiconductor device **100** which will be described later with reference to FIG. 4. The plurality of memory devices **2200a** may be fixed to one another by adhesive layers **2300** respectively at lower surfaces of the memory devices **2200a**, and may be connected to the package substrate **2100** by the wire **2400**. An external connection terminal **2800** may be at a lower surface of the package substrate **2100**.

(26) FIG. 4 is a cross-sectional view of a semiconductor device according to an exemplary embodiment. A semiconductor device **100** according to the exemplary embodiment of the disclosure may include flash memory such as 3D-NAND. In the specification, the semiconductor device **100** may be referred to as a “semiconductor storage device”.

(27) Referring to FIG. 4, the semiconductor device **100** may have a chip-to-chip (C2C) structure. The C2C structure may mean that an upper chip including a cell area CELL is fabricated on a first wafer, a lower chip including a peripheral circuit area is fabricated on a second wafer different from the first wafer, and the upper chip and the lower chip are subsequently connected to each other through a bonding method. In an implementation, the bonding method may mean a method of electrically connecting a bonding metal formed at an uppermost metal layer of the upper chip and a bonding metal formed at an uppermost metal layer of the lower chip. In an implementation, when the bonding metals are made of copper (Cu), the bonding method may be a Cu—Cu bonding method. The bonding metals may be made of aluminum or tungsten.

(28) The semiconductor device **100** may include a cell area CELL, a first peripheral circuit area PERI1, and a second peripheral circuit area PERI2. The cell area CELL may correspond to the cell area **1100S** described in conjunction with FIG. 1, and the first peripheral circuit area PERI1 and the second peripheral circuit area PERI2 may correspond to the peripheral circuit area **1100F** of FIG. 1.

(29) The cell area CELL may include a first substrate **102**, a cell area insulating layer **110**, stack insulating layers **120**, and gate electrodes **125**. The first substrate **102** may include a semiconductor material. In an implementation, the first substrate **102** may be a silicon substrate, a germanium substrate, a silicon germanium substrate, or a silicon-on-insulator (SOI) substrate. In an implementation, the first substrate **102** may include a group IV semiconductor, a group III-V compound semiconductor, or a group II-VI oxide semiconductor. As used herein, the term “or” is not an exclusive term, e.g., “A or B” would include A, B, or A and B.

(30) The stack insulating layers **120** and the gate electrodes **125** may constitute a memory stack. The gate electrodes **125** may extend (e.g., lengthwise) in a horizontal direction and may be spaced apart from one another in a vertical direction. The gate electrodes **125** may include the word lines

WL, the first upper line UL1, the second upper line UL2, the first lower line LL1 and the second lower line LL2, which are shown in FIG. 1. The stack insulating layers **120** may be between (e.g., vertically) adjacent ones of the gate electrodes **125**, respectively. The stack insulating layers **120** may also extend (e.g., lengthwise) in the horizontal direction and may be spaced apart from one another in the vertical direction. The stack insulating layers **120** may electrically insulate the gate electrodes **125** from one another.

(31) The cell area CELL may further include channel structures CH, cell contact plugs **130**, a through contact plug **132**, wiring layers **140**, contact plugs **142**, and first bonding pads **150**. Each channel structure CH may extend vertically through the memory stack. The channel structure CH may also partially extend through or into an upper surface of the first substrate **102**. The channel structure CH may correspond to one cell string CSTR of FIG. 1. The channel structure CH may have a column shape, and may have a tapered shape such that the lateral width of the channel structure CH is gradually reduced as the channel structure CH extends toward or is closer to the first substrate **102**.

(32) The cell contact plugs **130** may be connected to the gate electrodes **125**, respectively. The stack insulating layers **120** and the gate electrodes **125** may have a stepped structure. The cell contact plugs **130** may extend vertically through the cell area insulating layer **110** and, as such, may be connected to the stepped structure. The through contact plug **132** may extend vertically through the cell area insulating layer **110** such that the through contact plug **132** is connected to the first substrate **102**.

(33) The wiring layers **140** may be on the channel structures CH, the cell contact plugs **130**, and the through contact plug **132**, respectively. The channel structures CH, the cell contact plugs **130**, and the through contact plug **132** may be connected to the wiring layers **140** by the contact plugs **142**, respectively. The channel structures CH may be connected to the page buffer **1120** of the peripheral circuit area **1100F** of FIG. 1. The cell contact plugs **130** and the through contact plug **132** may be connected to the row decoder **1110** of the peripheral circuit area **1100F** of FIG. 1. The wiring layers **140** connected to the channel structures CH may correspond to the bit lines BL of FIG. 1, respectively.

(34) The first bonding pads **150** may be at a top portion of the cell area CELL. In an implementation, an upper surface (e.g., surface facing away from the first substrate **102** in the vertical direction) of each first bonding pad **150** may be coplanar with an upper surface of the cell area insulating layer **110**. The first bonding pads **150** may be connected to the first peripheral circuit area PERI1, and may be connected to corresponding ones of the wiring layers **140** via wiring contact plugs **152**, respectively. The first bonding pads **150** may include a dummy pad.

(35) The first peripheral circuit area PERI1 may be between the second peripheral circuit area PERI2 and the cell area CELL. The first peripheral circuit area PERI1 may include a second substrate **202**, a first peripheral area insulating layer **210**, first wiring layers **230**, second wiring layers **240**, third wiring layers **250**, and second bonding pads **260**. In an implementation, the second substrate **202** may include the same material as the first substrate **102**. A surface of the second substrate **202** in the first peripheral circuit area PERI1 may be referred to as a “first surface **203**”. A surface of the second substrate **202** opposite to the first surface **203** and in the second peripheral circuit area PERI2 may be referred to as a “second surface **204**”. The first peripheral area insulating layer **210** may cover the first surface **203**. A device isolation layer **212** and an impurity region **214** may be inside (e.g., may extend into the second substrate **202** at) the first surface **203**. A device **220** may be on the first surface **203** adjacent to the impurity region **214**. The device **220** may include an active device such as a transistor or a passive device such as an inductor, a resistor, or a capacitor.

(36) The first wiring layers **230**, the second wiring layers **240**, and the third wiring layers **250** may be in the first peripheral area insulating layer **210**. A corresponding one of the first wiring layers **230** may be connected to the impurity region **214** by a contact plug **232**. Each second wiring layer **240** may be under a corresponding one of the first wiring layers **230**, and may be connected to the

corresponding first wiring layer **230** by a first wiring contact plug **242**. Each third wiring layer **250** may be under a corresponding one of the second wiring layers **240**, and may be connected to the corresponding second wiring layers **240** by a second wiring contact plug **252**.

(37) Each second bonding pad **260** may be connected to a corresponding one of the third wiring layers **250** by a third wiring contact plug **262**, and may be under (e.g., at a bottom side of) the first peripheral area insulating layer **210**. In an implementation, a lower surface (e.g., first substrate **102**-facing surface) of each second bonding pad **260** may be coplanar with a lower surface of the first peripheral area insulating layer **210**. Each second bonding pad **260** may be bonded to a corresponding one of the first bonding pads **150**. In an implementation, each second bonding pad **260** may be connected to the corresponding first bonding pad **150** in a Cu—Cu bonding manner. The second bonding pads **260** may include a dummy pad. In an implementation, the dummy pad among the second bonding pads **260** may be connected to the dummy pad among the first bonding pads **150**.

(38) The second peripheral circuit area PERI2 may be on the first peripheral circuit area PERI1. The second peripheral circuit area PERI2 may include a second peripheral area insulating layer **310**, first wiring layers **340**, second wiring layers **350**, third wiring layers **360**, an upper insulating layer **370**, and an input/output pad **380**. As described above, the second surface **204** of the second substrate **202** may be in the second peripheral circuit area PERI2. A device isolation layer **312** and an impurity region **314** may be inside the second surface **204** (e.g., may extend into the second substrate **202** at the second surface **204** thereof). A device **320** may be on the second surface **204** adjacent to the impurity region **314**. In an implementation, a transistor of the device **320** in the second peripheral circuit area PERI2 may have a structure different from that of a transistor of the device **220** in the first peripheral circuit area PERI1. In an implementation, a gate electrode of the device **320** may include a material different from that of a gate electrode of the device **220**. The device **320** may include a gate dielectric layer with a structure and/or a material different from that of a gate dielectric layer of the device **320**. An operating voltage of the device **320** may differ from an operating voltage of the device **220**.

(39) The first wiring layers **340**, the second wiring layers **350**, and the third wiring layers **360** may be in the second peripheral area insulating layer **310**. A corresponding one of the first wiring layers **340** may be connected to the impurity region **314** by a contact plug **342**. Each second wiring layer **350** may be over a corresponding one of the first wiring layers **340**, and may be connected to the corresponding first wiring layer **340** by a first wiring contact plug **352**. Each third wiring layer **360** may be over a corresponding one of the second wiring layers **350**, and may be connected to the corresponding second wiring layers **350** by a second wiring contact plug **362**.

(40) The semiconductor device **100** according to the exemplary embodiment of the disclosure may further include a through electrode **330** extending vertically through the second substrate **202**, and a through electrode insulating layer **332** surrounding a side surface of the through electrode **330**. The through electrode **330** may extend from the second peripheral area insulating layer **310** in the second peripheral circuit area PERI2 through the second substrate **202**. In an implementation, an upper surface of the through electrode **330** may contact a corresponding one of the first wiring layers **340**. In an implementation, the upper surface of the through electrode **330** may contact a corresponding one of the second wiring layers **350** or a corresponding one of the third wiring layers **360**. In an implementation, a lower surface of the through electrode **330** may contact a through electrode plug **234**, and may be connected to a corresponding one of the first wiring layer **230** by the through electrode plug **234**. Accordingly, the second peripheral circuit area PERI2 may be electrically connected to the first peripheral circuit area PERI1 through the through electrode **330**, and may also be electrically connected to the cell area CELL through the first peripheral circuit area PERI1.

(41) The upper insulating layer **370** may be on the second peripheral area insulating layer **310**. The input/output pad **380** may be on the upper insulating layer **370**, and may be connected to a

corresponding one of the third wiring layers **360** through an input/output contact plug **382**. The input/output pad **380** may correspond to the chip pad **2210** of FIG. 2.

(42) As shown in FIG. 4, the semiconductor device **100** may include the first peripheral circuit area PERI1 on the first surface **203** of the second substrate **202** and the second peripheral circuit area PERI2 on the second surface **204** of the second substrate **202** and, as such, may realize a wider variety of wiring interconnections in the semiconductor device **100**, and may achieve a reduction in chip size in a horizontal direction.

(43) FIG. 5 is an enlarged view of the semiconductor device shown in FIG. 4. FIG. 5 shows upper and lower portions of a channel structure CH.

(44) Referring to FIG. 5, the channel structure CH may include an information storage layer **20**, a channel layer **30**, a buried insulating pattern **32**, and a conductive pad **34**. The channel layer **30** may be inside the information storage layer **20**. The buried insulating pattern **32** may be inside the channel layer **30**. The information storage layer **20** may include a tunnel insulating layer **22**, a charge storage layer **24**, and a blocking layer **26**. The charge storage layer **24** may be inside the blocking layer **26**. The tunnel insulating layer **22** may be inside the charge storage layer **24**. The conductive pad **34** may be at the upper portion of the channel structure CH, and may be connected to the channel layer **30**. In an implementation, the channel layer **30** may include polysilicon. The buried insulating pattern **32** may include silicon oxide, silicon nitride, silicon oxynitride, or a combination thereof. In an implementation, the blocking layer **26** and the tunnel insulating layer **22** may include silicon oxide, and the charge storage layer **24** may include silicon nitride.

(45) A connecting conductive layer **43** may be at or on an upper surface of a first substrate **102**, and may contact (e.g., directly contact) a side surface of the channel layer **30** while extending through the information storage layer **20**. A portion of the connecting conductive layer **43**, which contacts the channel layer **30**, may extend in a vertical direction. A supporter **44** may be on the connecting conductive layer **43**. The connecting conductive layer **43** and the supporter **44** may include polysilicon.

(46) FIGS. 6 to 16 are cross-sectional views of stages in a method of manufacturing a semiconductor device shown in FIG. 4 according to an exemplary embodiment.

(47) FIGS. 6 to 11 show formation of a structure corresponding to a cell area CELL. Referring to FIG. 6, a first substrate **102** and a cell area insulating layer **110** may be provided. The first substrate **102** may include a semiconductor material such as silicon. The cell area insulating layer **110** may be formed by patterning the first substrate **102**, and then filling an insulating material. The cell area insulating layer **110** may include silicon oxide, silicon nitride, silicon oxynitride, or a combination thereof.

(48) Stack insulating layers **120** and stack sacrificial layers **122** may be stacked on the first substrate **102** and the cell area insulating layer **110**. The stack insulating layers **120** may include a material having etch selectivity with respect to the stack sacrificial layers **122**. In an implementation, the stack insulating layers **120** may include silicon oxide, and the stack sacrificial layers **122** may include silicon nitride.

(49) Referring to FIG. 7, the stack insulating layers **120** and the stack sacrificial layers **122** may be trimmed. In an implementation, the stack insulating layers **120** and the stack sacrificial layers **122** may be trimmed to have a stepped structure, through repetition of a photolithography process and an etching process. An insulating material may further be deposited such that the cell area insulating layer **110** covers the stack insulating layers **120** and the stack sacrificial layers **122**.

(50) Referring to FIG. 8, channel structures CH extending vertically through the stack insulating layers **120** and the stack sacrificial layers **122** may be formed. In an implementation, channel holes extending vertically through the stack insulating layers **120** and the stack sacrificial layers **122** may be formed, and the channel holes may be filled with a channel material, thereby forming the channel structures CH. Each channel structure CH may have a column shape, and may have a tapered shape such that the lateral width of the channel structure CH is gradually reduced as the

channel structure CH extends toward the first substrate **102**.

(51) Referring to FIG. **9**, the stack sacrificial layers **122** may be substituted by gate electrodes **125**. In an implementation, a word line cut extending vertically through the stack insulating layers **120** and the stack sacrificial layers **122**, and the stack sacrificial layers **122** may be selectively removed through the word line cut. The gate electrodes **125** may be filled in spaces formed through removal of the stack sacrificial layers **122**, respectively. In an implementation, the gate electrodes **125** may include a metal such as tungsten, polysilicon, or a metal silicide material. The stack insulating layers **120** and the gate electrodes **125** may constitute a memory stack.

(52) Referring to FIG. **10**, cell contact plugs **130** and a through contact plug **132** extending vertically through the cell area insulating layer **110** may be formed. The cell contact plugs **130** and the through contact plug **132** may be formed by forming contact holes extending vertically through the cell area insulating layer **110**, and then depositing a conductive material in the contact holes. The cell contact plugs **130** may be connected to the gate electrodes **125**, respectively, and the through contact plug **132** may be connected to the first substrate **102**.

(53) Referring to FIG. **11**, wiring layers **140**, contact plugs **142**, first bonding pads **150**, and wiring contact plugs **152** may be formed on the resultant structure of FIG. **10**. The wiring layers **140**, contact plugs **142**, first bonding pads **150**, and wiring contact plugs **152** may be formed by repeating processes of forming an insulating material layer on the resultant structure of FIG. **10**, partially etching the insulating layer through a patterning process, and depositing a conductive material.

(54) The wiring layers **140** may be connected to corresponding ones of the channel structures CH, the cell contact plugs **130** and the through contact plug **132**, respectively. The first bonding pads **150** may be at a top portion of the cell area insulating layer **110**. After formation of the first bonding pads **150**, a planarization process may be performed. An upper surface of each first bonding pad **150** may not be covered by the cell area insulating layer **110**.

(55) Referring to FIG. **12**, a structure corresponding to a first peripheral circuit area PERI1 may be formed. First, a device isolation layer **212** and an impurity region **214** may be formed in a second substrate **202**, and a device **220** may be formed on the second substrate **202**. The device isolation layer **212** may be formed by depositing silicon oxide in the second substrate **202**, and the impurity region **214** may be formed through an ion implementation process.

(56) First wiring layers **230**, contact plugs **232**, a through electrode plug **234**, second wiring layers **240**, first wiring contact plugs **242**, third wiring layers **250**, second wiring contact plugs **252**, second bonding pads **260** and third wiring contact plugs **262** may be formed on the second substrate **202**. The contact plugs **232**, the through electrode plug **234**, the first wiring contact plugs **242**, the second wiring contact plugs **252** and the third wiring contact plugs **262** may extend vertically between corresponding ones of the second substrate **202**, the first wiring layers **230**, the second wiring layers **240** and the third wiring layers **250**. The first wiring layers **230**, the contact plugs **232**, the through electrode plug **234**, the second wiring layers **240**, the first wiring contact plugs **242**, the third wiring layers **250**, the second wiring contact plugs **252**, the second bonding pads **260**, and the third wiring contact plugs **262** may be formed by repeating processes of forming an insulating material layer on the second substrate **202**, patterning the insulating material layer, and depositing a conductive material.

(57) Referring to FIG. **13**, the structure corresponding to the first peripheral circuit area PERI1 may be bonded to the structure corresponding to the cell area CELL through pressing. The structure corresponding to the first peripheral circuit area PERI1 may be bonded in an inverted state in which the second bonding pads **260** are directed downwards (e.g., toward the cell area CELL). The structures respectively corresponding to the first peripheral circuit area PERI1 and the cell area CELL may be bonded to each other without a separate adhesive therebetween. The first bonding pads **150** may be bonded to corresponding ones of the second bonding pads **260**, respectively, and the cell area insulating layer **110** may be bonded to a first peripheral area insulating layer **210**. A

surface treatment process such as hydrogen plasma treatment may further be performed before the bonding process.

(58) Referring to FIG. 14, an upper surface of the second substrate **202** may be partially etched. The etching process may be a process of planarizing the upper surface of the second substrate **202** and, as such, the thickness of the second substrate **202** may be reduced. In an implementation, the thickness of the etched second substrate **202** may be 300 nm to 3 μm .

(59) FIGS. 15 and 16 show formation of a structure corresponding to a second peripheral circuit area PERI2. As noted above, the surface of the second substrate **202** in the first peripheral circuit area PERI1 may be referred to as the “first surface **203**”, and a surface of the second substrate **202** opposite to the first surface **203** and in the second peripheral circuit area PERI2 may be referred to as the “second surface **204**”.

(60) Referring to FIG. 15, a device isolation layer **312** and an impurity region **314** may be formed inside the second surface **204**, and a device **320** and a second peripheral area insulating layer **310** may be formed on the second surface **204**. Thereafter, contact plugs **342** and a through electrode **330**, which extend vertically through the second peripheral area insulating layer **310**, may be formed. The contact plugs **342** and the through electrode **330** may be formed by anisotropically etching the second peripheral area insulating layer **310**, and depositing a conductive material. The through electrode **330** may be formed to extend through the second substrate **202** such that the through electrode **330** contacts the through electrode plug **234**. The through electrode **330** may have a column shape, and may have a tapered shape such that the lateral width of the through electrode **330** is gradually reduced as the through electrode **330** extends from the second surface **204** to the first surface **203**. In an implementation, a through electrode insulating layer **332** may be formed before formation of the through electrode **330** in order to electrically insulate the second substrate **202** and the through electrode **330** from each other.

(61) First wiring layers **340** may be formed on the second peripheral area insulating layer **310** such that the first wiring layers **340** are connected to corresponding ones of the contact plugs **342** and the through electrode **330**, respectively.

(62) Referring to FIG. 16, second wiring layers **350**, first wiring contact plugs **352**, third wiring layers **360**, and second wiring contact plugs **362** may be formed on the resultant structure of FIG. 15. The second wiring layers **350**, the first wiring contact plugs **352**, the third wiring layers **360**, and the second wiring contact plugs **362** may be formed by repeating processes of forming an insulating material layer on the resultant structure of FIG. 15, patterning the insulating material layer, and depositing a conductive material.

(63) Referring back to FIG. 4, an upper insulating layer **370**, an input/output pad **380** and an input/output contact plug **382** may be formed. The upper insulating layer **370** may be formed by depositing an insulating material on the second peripheral area insulating layer **310**. The input/output contact plug **382** may be formed by anisotropically etching the second peripheral area insulating layer **310** and the upper insulating layer **370**, and depositing a conductive material. The input/output pad **380** may be formed on the input/output contact plug **382**.

(64) FIG. 17 is a cross-sectional view of a semiconductor device according to an exemplary embodiment. Detailed descriptions of components that are the same as or similar to those of the semiconductor device **100** of FIG. 4 may be omitted.

(65) Referring to FIG. 17, a semiconductor device **100a** may include a through electrode **330a** extending vertically through a second substrate **202** and electrically connecting a first peripheral circuit area PERI1 and a second peripheral circuit area PERI2 to each other, and a through electrode insulating layer **332a** surrounding a side surface of the through electrode **330a**. The through electrode **330a** may have a tapered shape such that the lateral width of the through electrode **330a** is gradually reduced as the through electrode **330a** extends from the first surface **203** to the second surface **204**. An upper surface of the through electrode **330a** may contact a through electrode plug **342a** connected to a first wiring layer **340**. A lower surface of the through

electrode **330a** may contact a through electrode plug **234** connected to a first wiring layer **230**. The upper surface of the through electrode **330a** may be substantially coplanar with the second surface **204**, whereas the lower surface of the through electrode **330a** may be substantially coplanar with the first surface **203**.

(66) FIGS. **18** to **21** are cross-sectional views of stages in a method of manufacturing a semiconductor device shown in FIG. **17** according to an exemplary example embodiment.

(67) Referring to FIG. **18**, a structure corresponding to a first peripheral circuit area PERI1 may be formed. A through electrode **330a** extending vertically in a second substrate in the first peripheral circuit area PERI1 and a through electrode insulating layer **332a** surrounding a side surface of the through electrode **330a** may be formed. The through electrode **330a** and the through electrode insulating layer **332a** may be formed by anisotropically etching the second substrate **202**, and depositing an insulating material and a conductive material.

(68) Referring to FIG. **19**, the structure corresponding to the first peripheral circuit area PERI1 may be bonded to a structure corresponding to a cell area CELL.

(69) Referring to FIG. **20**, after bonding of the structure corresponding to the first peripheral circuit area PERI1, an upper surface of the second substrate **202** opposite to a first surface **203** may be etched such that the through electrode **330a** is partially exposed. The process of etching the second substrate **202** may include a planarization process and an etch-back process. In an implementation, the thickness of the etched second substrate **202** may be 300 nm to 3 μ m.

(70) Referring to FIG. **21**, the exposed through electrode **330a** may be planarized. A second peripheral area insulating layer **310**, a device isolation layer **212**, an impurity region **214**, a transistor, first wiring layers **230**, and contact plugs may be formed on a second surface **204** of the second substrate **202**.

(71) Again referring to FIG. **17**, the first wiring layers **230**, second wiring layers **240**, third wiring layers **250**, an input/output pad **380**, and plugs connecting corresponding ones of the first wiring layers **230**, the second wiring layers **240**, the third wiring layers **250**, and the input/output pad **380** may be formed and, as such, a second peripheral circuit area PERI2 may be formed.

(72) FIGS. **22** to **29** are cross-sectional views of semiconductor devices according to exemplary embodiments.

(73) Referring to FIG. **22**, a semiconductor device **100b** may include a through electrode **330b** extending vertically through a second substrate **202** and electrically connecting a first peripheral circuit area PERI1 and a second peripheral circuit area PERI2 to each other, and a through electrode insulating layer **332b** surrounding a side surface of the through electrode **330b**. In an implementation, the through electrode **330b** may extend partially through a first peripheral area insulating layer **210** and, as such, may be connected to a first wiring layer **230** of the first peripheral circuit area PERI1. An upper surface of the through electrode **330b** may contact a first wiring layer **340**, and a lower surface of the through electrode **330b** may contact the first wiring layer **230**.

(74) Referring to FIG. **23**, a semiconductor device **100c** may include a lower insulating layer **370c**, an input/output pad **380c**, and an input/output contact plug **382c**. In an implementation, the input/output pad **380c** may be in the cell area CELL. In an implementation, the lower insulating layer **370c** may be under the first substrate **102**, and the input/output pad **380c** may be under the lower insulating layer **370c**. The input/output pad **380c** may be connected to a first bonding pad **150** by the input/output contact plug **382c** which extends through the cell area insulating layer **110** and the lower insulating layer **370c**. In an implementation, the semiconductor device **100c** may only include the input/output pad **380c**, as illustrated in FIG. **23**. In an implementation, the semiconductor device **100c** may include both the input/output pad **380c** and the input/output pad **380** of FIG. **4**.

(75) Referring to FIG. **24**, a semiconductor device **100d** may include a second lower substrate **202d**, a buried insulating layer **301d**, and a second upper substrate **302d**. Referring to FIGS. **14** to **16**, in an embodiment, after bonding of the first peripheral circuit area PERI1, the second substrate

202 is etched, and an insulating material and a silicon substrate may then be deposited. The structure corresponding to the second peripheral circuit area **PERI2** may be formed on the silicon substrate. In an implementation, the second lower substrate **202d** may include the same material as the second substrate **202**. The buried insulating layer **301d** may include silicon oxide, silicon nitride, silicon oxynitride, or a combination thereof. The second upper substrate **302d** may include polysilicon.

(76) Referring to FIG. 25, a semiconductor device **100e** may include a cell area **CELL**, a first peripheral circuit area **PERI1** under the cell area **CELL**, and a second peripheral circuit area **PERI2** over the cell area **CELL** (e.g., on an opposite side relative to the first peripheral circuit area **PERI1**). The cell area **CELL** and the first peripheral circuit area **PERI1** may include configurations identical or similar to those of the cell area **CELL** and the first peripheral circuit area **PERI1** in FIG. 4. In an implementation, wiring structures in the cell area **CELL** and the first peripheral circuit area **PERI1** in FIG. 25 may be partially different from those of the cell area **CELL** and the first peripheral circuit area **PERI1** in FIG. 4.

(77) The cell area **CELL** may be connected to the first peripheral circuit area **PERI1** in an inverted state in which first bonding pads **150** are directed downwards. In an implementation, the cell area **CELL** may include first bonding pads **150** under a cell area insulating layer **110**, and the first peripheral circuit area **PERI1** may include second bonding pads **260** bonded to the first bonding pads **150**, respectively, and may be on a first peripheral area insulating layer **210**. A surface of a first substrate **102** in the cell area **CELL** may be referred to as a “first surface **103**”, and a surface of the first substrate **102** opposite to the first surface **103** may be referred to as a “second surface **104**”.

(78) The cell area **CELL** may include cell contact plugs **130**, a through contact plug **132e**, and connecting wiring layers **140e**. The cell contact plugs **130** may be connected to gate electrodes **125**, respectively. The through contact plug **132e** may be connected to the first substrate **102**. The cell contact plugs **130** and the through contact plug **132e** may be connected to the connecting wiring layers **140e** via contact plugs **142**, respectively.

(79) A structure corresponding to the second peripheral circuit area **PERI2** may be formed at the first substrate **102** of the cell area **CELL** after a structure corresponding to the cell area **CELL** is bonded to a structure corresponding to the first peripheral circuit area **PERI1**. In an implementation, the second peripheral circuit area **PERI2** may be formed on the second surface **104** of the first substrate **102**. The second peripheral circuit area **PERI2** may include a device isolation layer **312** and an impurity region **314**, which are inside the second surface **104**, a device **320** on the second surface **104**, first wiring layers **340**, second wiring layers **350**, third wiring layers **360**, and an input/output pad **380**. The second peripheral circuit area **PERI2** may also include contact plugs **342**, first wiring contact plugs **352**, second wiring contact plugs **362**, and an input/output contact plug **382**, which connect corresponding ones of the first wiring layers **340**, the second wiring layers **350**, the third wiring layers **360**, and the input/output pad **380**, respectively.

(80) The second peripheral circuit area **PERI2** may further include a through electrode **330** connecting the second peripheral circuit area **PERI2** and the cell area **CELL**. The through electrode **330** may have a tapered shape such that the lateral width of the through electrode **330** is gradually reduced as the through electrode **330** extends from the second surface **104** to the first surface **103**. The through electrode **330** may electrically connect the second peripheral circuit area **PERI2** to the cell area **CELL**. In an implementation, the through electrode **330** may vertically extend from a corresponding one of the first wiring layers **340** through the second peripheral area insulating layer **310** and the first substrate **102** such that the through electrode **330** contacts the through contact plug **132e**. The through contact plug **132e** may further be connected to corresponding ones of the connecting wiring layers **140e** and the first bonding pads **150**. Accordingly, the through contact plug **132e** may electrically connect the second peripheral circuit area **PERI2** to the first peripheral circuit area **PERI1**.

(81) Referring to FIG. 26, a semiconductor device **100f** may include a through electrode **330f** extending vertically through a second substrate **202** and electrically connecting a first peripheral circuit area PERI1 and a second peripheral circuit area PERI2 to each other, and a through electrode insulating layer **332f** surrounding a side surface of the through electrode **330f**. The through electrode **330f** may have a tapered shape such that the lateral width of the through electrode **330f** is gradually reduced as the through electrode **330f** extends from a first surface **103** to a second surface **104**. An upper surface of the through electrode **330f** may contact a through electrode plug **342f** connected to a first wiring layer **340**. A lower surface of the through electrode **330f** may contact a through contact plug **132e**. The upper surface of the through electrode **330f** may be substantially coplanar with the second surface **104**, and the lower surface of the through electrode **330f** may be substantially coplanar with the first surface **103**.

(82) Referring to FIG. 27, a semiconductor device **100g** may include a first lower substrate **102g**, a buried insulating layer **301g**, and a first upper substrate **320g**. In an implementation, the first lower substrate **102g** may include a semiconductor material such as monocrystalline silicon. The buried insulating layer **301g** may include silicon oxide, silicon nitride, silicon oxynitride, or a combination thereof. The first upper substrate **302g** may include polysilicon.

(83) Referring to FIG. 28, a semiconductor device **100h** may include a cell area CELL, a first peripheral circuit area PERI1 on the cell area CELL, and a second peripheral circuit area PERI2 on the first peripheral circuit area PERI1 and a third peripheral circuit area PERI3 under the cell area CELL. The first peripheral circuit area PERI1 and the second peripheral circuit area PERI2 may include configurations similar to those of the semiconductor device **100** of FIG. 4.

(84) In an implementation, the cell area CELL and the third peripheral circuit area PERI3 may have a cell-on-peri (COP) structure. The third peripheral circuit area PERI3 may include a third substrate **402**, a device **420** on the third substrate **402**, and wiring layers **430** connected to the device **420**. The cell area CELL may include a first through contact plug **440** and a second through contact plug **450** which are connected to the wiring layers **430**.

(85) The first through contact plug **440** may extend vertically through the first substrate **102** such that the first through contact plug **440** contacts a contact plug **142**. The first through contact plug **440** may be electrically insulated from the first substrate **102** by a buried insulating layer **104h** in the first substrate **102**. The first through contact plug **440** may also be electrically insulated from gate electrodes **125** by a contact insulating layer **122h**. In an implementation, the contact insulating layer **122h** may extend vertically through the gate electrodes **125**, and the first through contact plug **440** may extend vertically through the first substrate **102** and the contact insulating layer **122h** such that the first through contact plug **440** is electrically connected to the third peripheral circuit area PERI3.

(86) The second through contact plug **450** may extend vertically from a corresponding one of the wiring layers **430** through a cell area insulating layer **110** such that the second through contact plug **450** contacts another contact plug **142**. The second through contact plug **450** may electrically connect the third peripheral circuit area PERI3 to the first peripheral circuit area PERI1 through a first bonding pad **150** and a second bonding pad **260**.

(87) Referring to FIG. 29, a semiconductor device **100i** may include a first cell area CELL1, a first peripheral circuit area PERI1 on the first cell area CELL1, a second cell area CELL2 on the first peripheral circuit area PERI1 and a second peripheral circuit area PERI2 on the second cell area CELL2.

(88) The first cell area CELL1 and the first peripheral circuit area PERI1 may include configurations identical or similar to the cell area CELL and the first peripheral circuit area PERI1 in FIG. 4, respectively. In an implementation, wiring structures of the first cell area CELL1 and the first peripheral circuit area PERI1 in FIG. 29 may be partially different from those of the cell area CELL and the first peripheral circuit area PERI1 in FIG. 4. In an implementation, the first peripheral circuit area PERI1 may include a through electrode **230i** extending through a second

substrate **202**, a through electrode insulating layer **232i** surrounding a side surface of the through electrode **230i**, and a third bonding pad **260i**. An upper surface of the through electrode **230i** may contact the third bonding pad **260i**. The through electrode **230i** may have a tapered shape such that the lateral width of the through electrode **230i** is gradually reduced as the through electrode **230i** extends toward the third bonding pad **260i**.

(89) The second cell area **CELL2** and the second peripheral circuit area **PERI2** may include configurations identical or similar to those of the cell area **CELL** and the second peripheral circuit area **PERI2** in FIG. **25**, respectively. In an implementation, wiring structures of the second cell area **CELL2** and the second peripheral circuit area **PERI2** may be partially different from those of the cell area **CELL** and the second peripheral circuit area **PERI2** in FIG. **25**. The second cell area **CELL2** may include a fourth bonding pad **150** bonded to the third bonding pad **260i**.

(90) One or more embodiments may provide a semiconductor device having peripheral circuit areas on both surfaces of a substrate.

(91) In accordance with the exemplary embodiments of the disclosure, peripheral circuit areas may be at opposite surfaces of a substrate and, as such, a wider variety of wiring interconnections may be embodied in a semiconductor device.

(92) Example embodiments have been disclosed herein, and although specific terms are employed, they are used and are to be interpreted in a generic and descriptive sense only and not for purpose of limitation. In some instances, as would be apparent to one of ordinary skill in the art as of the filing of the present application, features, characteristics, and/or elements described in connection with a particular embodiment may be used singly or in combination with features, characteristics, and/or elements described in connection with other embodiments unless otherwise specifically indicated. Accordingly, it will be understood by those of skill in the art that various changes in form and details may be made without departing from the spirit and scope of the present invention as set forth in the following claims.

Claims

1. A semiconductor device, comprising: a cell area including a first substrate, gate electrodes on the first substrate and spaced apart from one another in a vertical direction, a channel structure extending vertically through the gate electrodes, cell contact plugs connected to the gate electrodes and extending in the vertical direction, a through contact plug connected to the first substrate and extending in the vertical direction, and first bonding pads electrically connected to the channel structure, the cell contact plugs and the through contact plug; a first peripheral circuit area connected to the cell area on the cell area, the first peripheral circuit area including second bonding pads on the first bonding pads; a second peripheral circuit area connected to the first peripheral circuit area and comprising a logic circuit configured to control operations of the semiconductor device; and a second substrate including a lower surface in the first peripheral circuit area and an upper surface in the second peripheral circuit area and opposite to the lower surface of the second substrate, wherein the first peripheral circuit area includes: a first device isolation layer extending from the lower surface of the second substrate toward the upper surface of the second substrate; and a first impurity region formed in the lower surface of the second substrate and adjacent the first device isolation layer, and wherein the second peripheral circuit area includes: a device directly disposed on the upper surface of the second substrate, a through electrode extending vertically through the second substrate and connected to the first peripheral circuit area, a second device isolation layer extending from the upper surface of the second substrate toward the lower surface of the second substrate, and a second impurity region formed in the upper surface of the second substrate and adjacent the second device isolation layer.

2. The semiconductor device as claimed in claim 1, wherein the through electrode is electrically connected to the cell area through at least one of the second bonding pads and at least one of the

first bonding pads.

3. The semiconductor device as claimed in claim 1, wherein: the first peripheral circuit area includes: a wiring layer electrically connected to one of the second bonding pads, and a first through electrode plug connected to the wiring layer, and a lower surface of the through electrode contacts the first through electrode plug.
4. The semiconductor device as claimed in claim 1, wherein: the second peripheral circuit area further includes an input/output pad on an upper surface thereof, and the input/output pad is electrically connected to the device in the second peripheral circuit area.
5. The semiconductor device as claimed in claim 1, wherein a lower surface of the through electrode is substantially coplanar with the lower surface of the second substrate.
6. The semiconductor device as claimed in claim 1, further comprising a through electrode insulating layer surrounding a side surface of the through electrode.
7. The semiconductor device as claimed in claim 1, wherein the through electrode has a tapered shape such that a width of the through electrode is gradually reduced as the through electrode extends from the upper surface of the second substrate to the lower surface of the second substrate.
8. The semiconductor device as claimed in claim 1, wherein the through electrode has a tapered shape such that a width of the through electrode is gradually reduced as the through electrode extends from the lower surface of the second substrate to the upper surface of the second substrate.
9. The semiconductor device as claimed in claim 8, wherein: the second peripheral circuit area includes: a wiring layer electrically connected to the device, and a second through electrode plug connected to the wiring layer, and an upper surface of the through electrode contacts the second through electrode plug.
10. The semiconductor device as claimed in claim 8, wherein an upper surface of the through electrode is substantially coplanar with the upper surface of the second substrate.
11. The semiconductor device as claimed in claim 1, wherein: the first peripheral circuit area includes a wiring layer electrically connected to one of the second bonding pads, and a lower surface of the through electrode is at a lower level than the lower surface of the second substrate and contacting the wiring layer.
12. The semiconductor device as claimed in claim 1, wherein: the first peripheral circuit area further includes an input/output pad at a lower surface thereof, and the input/output pad is electrically connected to one of the first bonding pads.
13. The semiconductor device as claimed in claim 1, wherein the second substrate includes: a second lower substrate, a second upper substrate on the second lower substrate, and a buried insulating layer between the second lower substrate and the second upper substrate.
14. The semiconductor device as claimed in claim 1, further comprising a third peripheral circuit area under the cell area and connected to the cell area, wherein the cell area includes: a contact insulating layer extending vertically through the gate electrodes, and a through contact plug extending vertically through the first substrate and the contact insulating layer such that the through contact plug is connected to the third peripheral circuit area.
15. A semiconductor device, comprising: a cell area including a first substrate, gate electrodes on a lower surface of the first substrate and spaced apart from one another in a vertical direction, a channel structure extending vertically through the gate electrodes, cell contact plugs connected to the gate electrodes and extending in the vertical direction, a through contact plug connected to the first substrate and extending in the vertical direction, and first bonding pads electrically connected to the channel structure, the cell contact plugs, and the through contact plug; a first peripheral circuit area under the cell area and connected to the cell area, the first peripheral circuit area including second bonding pads on the first bonding pads; and a second peripheral circuit area connected to the cell area on the cell area and comprising a logic circuit configured to control operations of the semiconductor device, wherein the lower surface of the first substrate is in the cell area, and an upper surface of the first substrate is in the second peripheral circuit area and opposite

to the lower surface of the first substrate, and wherein the second peripheral circuit area includes: an active device directly disposed on the upper surface of the first substrate, a device isolation layer extending from the upper surface of the first substrate toward the lower surface of the first substrate, an impurity region formed in the upper surface of the first substrate and adjacent the device isolation layer, and a through electrode extending vertically through the first substrate such that the through electrode is electrically connected to the cell area, wherein a horizontal width of an upper surface of the device isolation layer is greater than a horizontal width of a lower surface of the device isolation layer.

16. The semiconductor device as claimed in claim 15, wherein the through electrode is electrically connected to the first peripheral circuit area through the through contact plug and one of the first bonding pads.

17. The semiconductor device as claimed in claim 15, wherein the through electrode has a tapered shape such that a width of the through electrode is gradually reduced as the through electrode extends from the lower surface of the first substrate to the upper surface of the first substrate.

18. The semiconductor device as claimed in claim 15, wherein the first substrate includes a first lower substrate, a first upper substrate on the first lower substrate, and a buried insulating layer between the first lower substrate and the first upper substrate.

19. A semiconductor device, comprising: a cell area including a first substrate, gate electrodes on the first substrate and spaced apart from one another in a vertical direction, a channel structure extending vertically through the gate electrodes, cell contact plugs connected to the gate electrodes and extending in the vertical direction, a through contact plug connected to the first substrate and extending in the vertical direction, and first bonding pads electrically connected to the channel structure, the cell contact plugs, and the through contact plug; a first peripheral circuit area on the cell area and connected to the cell area, the first peripheral circuit area including second bonding pads on the first bonding pads, respectively; a second peripheral circuit area on the first peripheral circuit area and connected to the first peripheral circuit area and comprising a logic circuit configured to control operations of the semiconductor device; a semiconductor storage device including a second substrate including a lower surface in the first peripheral circuit area and an upper surface in the second peripheral circuit area, and an input/output pad electrically connected to at least one of the first peripheral circuit area or the second peripheral circuit area; and a controller electrically connected to the semiconductor storage device through the input/output pad such that the controller controls the semiconductor storage device, wherein the first peripheral circuit area includes: a first device isolation layer extending from the lower surface of the second substrate toward the upper surface of the second substrate; and a first impurity region formed in the lower surface of the second substrate and adjacent the first device isolation layer, and wherein the second peripheral circuit area includes: a device directly on the upper surface of the second substrate, a through electrode extending vertically through the second substrate such that the through electrode is electrically connected to the first peripheral circuit area, a second device isolation layer extending from the upper surface of the second substrate toward the lower surface of the second substrate, and a second impurity region formed in the upper surface of the second substrate and adjacent the second device isolation layer.

20. The semiconductor device as claimed in claim 19, wherein the through electrode is electrically connected to the cell area through at least one of the second bonding pads and at least one of the first bonding pads.
